STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-CURVE-ISOLATED-LCI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3072-3081 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3083-3100, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 34 of queue_examples-defos.json (pb=816, stmt 3072-3081, proof 3083-3100).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DY7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1425–1440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYX: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1600–1608. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZN: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2971–2988. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZH: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2953–2957. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZY: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3100–3100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A25: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3100–3100. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0DY7 let X be a scheme over k. Assume*

(1) *X is separated, finite type over k and $\dim(X) \leq 1$,*
(2) *X is a local complete intersection over k, and*
(3) *$X \to \mathrm{Spec}(k)$ is smooth except at finitely many points.*

Then $\mathcal{D}ef_X$ is unobstructed.

Proof. Let $p_1, \ldots, p_n \in X$ be the points where $X \to \mathrm{Spec}(k)$ isn't smooth. Choose an affine open $U_2 \subset X$ containing $p_1, \ldots, p_n$ and the generic point of every irreducible component of X. This is possible by Varieties, Lemma 0A25 and Properties, Lemma 01ZY. Then $X \setminus U_2$ is finite and we can choose an affine open $U_1 \subset X \setminus \{p_1, \ldots, p_n\}$ such that $X = U_1 \cup U_2$. Set $U_{12} = U_1 \cap U_2$. Then U_1 and U_{12} are smooth affine schemes over k. We conclude that

$$\mathcal{D}ef_X \longrightarrow \mathcal{D}ef_{U_2}$$

is smooth by Lemma 0DZN. Applying Lemmas 0DYX and 0DZH we win. □